# SMART INDIA HACKATHON 2024

# InFlux

## Charging ahead with seamless power

- **Problem Statement ID- SIH1595**
- **Problem Statement Title- InFlux**
- **Theme- Smart Vehicle**
- **PS Category- Software**
- **Team ID- MITADTSW337**
- **Team Name- C2 Innovative**

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# InFlux

## Charging ahead with seamless power

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## ❖Proposed Solution

- Provide a centralized website for locating EV charging stations.
- Collaborate with all major EV charger brands.
- Offer a unified wallet for easy payments across different chargers.

# C2 Innovatives

## SOFTWARES

### FRONTEND

- React.js
- HTML / CSS
- BootStrap

### BACKEND

- Node.js
- Express.js
- API Calls ( Axios, Fetch )
- Google Maps API
- Payment Gateway API

### DATABASES

- MONGODB
- MY SQL

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## ROADMAP

### DEFINING OBJECTIVES

The objective is to simplify **EV charger discovery** and payments through a **unified wallet** with brand collaborations.

### PLANNING

The planning phase involves designing the prototype, researching APIs for maps, authentication and payment gateways, and setting milestones for development.

### DEVELOPMENT

Development includes building the backend, creating a user-friendly interface, and integrating unified payment systems and route mapping.

### TESTING

Testing includes verifying website functionality, payment integration, GPS mapping accuracy, and ensuring overall performance and bug fixes

### LAUNCH

Launch involves deploying the website, promoting brand collaborations, and monitoring performance to address any post-launch issues.

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# FEASIBILITY AND VIABILITY

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Challenge 1
**Demand vs. Infrastructure**

Rapid EV adoption may outpace charging infrastructure development.

**Solution**

- **Growing demand** for **web-based EV charging solutions** with market growth.

Challenge 2
**Brand Integration:**

Managing multiple charger APIs can lead to compatibility and data consistency issues.

**Solution**

- **Collaborating with brands** reduces infrastructure challenges.

Challenge 3
**Payment Security:**

Users may hesitate due to concerns about online payment safety.

**Solution**

- **Secure payment systems** ensures user trust and adoptions.

Challenge 4
**Scalability:**

Ensuring smooth platform growth and future mobile compatibility can be challenging.

**Solution**

- Scalable for future additions like **mobile compatibility.**

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# IMPACT AND BENEFITS

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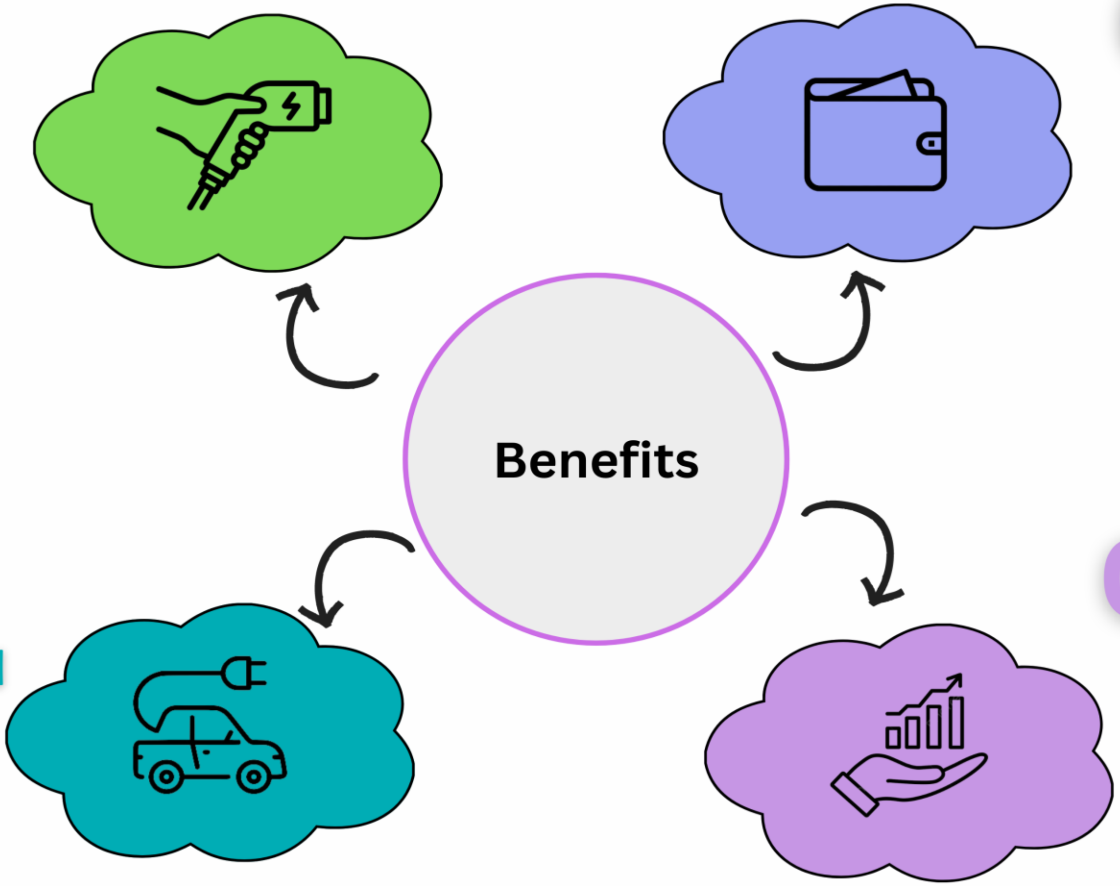


## 01 EFFICIENT CHARGER DISCOVERY

The website simplifies locating EV charging stations, providing users with real-time station data.

## 02 ENHANCED PAYMENT CONVENIENCE

The unified wallet streamlines payments, boosting user convenience and accelerating transaction speed.

## 03 PROMOTING SUSTAINABLE TRANSPORTATION

By supporting EV adoption, the platform contributes to the growth of green transportation and environmental sustainability.

## 04 REVENUE GROWTH FOR CHARGER PROVIDERS

Collaboration with EV charger brands creates new revenue streams and business opportunities for providers.

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# RESEARCH AND REFERENCES

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